Statistical Machine Learning Analysis Report

Bank Marketing — Subscription Prediction, Resampling & Model Selection

Tahar Masmaliyev & Abdulla Akhundzade

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1. Dataset Overview

The **UCI Bank Marketing** dataset (Moro et al., 2011) contains 4,521 telephone marketing records from a Portuguese bank (2008–2010). After one-hot encoding, 43 features predict whether a client subscribes to a term deposit. The subscription rate is **11.52%** (severe 88:12 class imbalance). An 80/20 stratified train-test split is used. `duration` (call length) is the strongest numeric correlate with the target (r = 0.40).

2. Prior Classification & Regression Results

Model	Accuracy	Precision	Recall	F1	ROC AUC
Logistic Regression	0.892	0.554	0.298	0.388	**0.891**
LDA	0.881	0.477	0.394	0.432	0.889
QDA	0.845	0.350	0.404	0.375	0.820
Naive Bayes	0.825	0.318	0.452	0.373	0.786

Logistic Regression leads (AUC = 0.891). `duration` is the dominant predictor (coef = 1.121, Z = 18.69, p $\approx$ 0, OR = 3.07), followed by `poutcome_success` and `contact_unknown`. Also significant: `loan` (p = 0.002), `housing` (p = 0.032), `campaign` (p = 0.027). Non-significant: `age` (p = 0.390), `balance` (p = 0.504), `pdays` (p = 0.932).

LDA (0.889) nearly matches LR; **QDA** (0.820) underperforms, confirming shared covariance holds. **NB** (0.786) has lowest AUC but highest recall (0.452), with class priors 0.885/0.115. Confusion matrices: LDA (756 TN, 45 FP, 63 FN, 41 TP); QDA (723 TN, 78 FP, 62 FN, 42 TP); NB (700 TN, 101 FP, 57 FN, 47 TP).

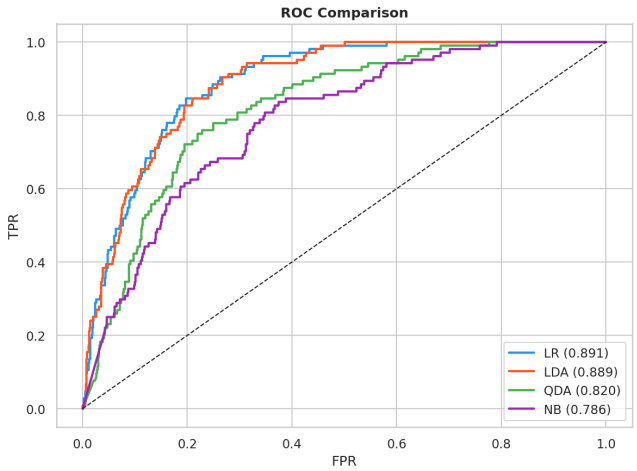

Figure 1: ROC curves. LR (0.891) and LDA (0.889) lead; QDA (0.820) and NB (0.786) trail.

Optimal LDA threshold: Youden's J = **0.027** (far below 0.5), yielding TPR ≈ 0.90, FPR ≈ 0.27. **Confounding:** `previous` changed `duration` coefficient by only **0.07%** — not a confounder.

Regression (Linear vs Poisson): Both Linear (RMSE = 2.599, R^2 = –0.025) and Poisson (RMSE = 2.600, R^2 = –0.026) yield negative R^2: features do not explain `campaign` counts. Poisson is theoretically preferred (non-negativity) but neither is practically useful.

3. Task 1 — Resampling Methods

Task 1a: Cross-Validation (K=5, K=10, LOOCV)

Polynomial degrees 1–3 evaluated via K=5, K=10, and LOOCV (n=250). All methods agree: **degree 1** achieves minimum MSE. Degree 3 shows massive error increase, especially LOOCV (highest variance). K=10 provides the best bias-variance tradeoff.

Task 1b: Bootstrap

The .632 estimator corroborates CV. For degree 1, mean OOB MSE = 12,631,525, 95% CI [7,036,284, 20,439,447]. Bootstrap and CV K=5 both select degree 1; degree 3 shows extreme variance inflation.

Task 1c: MSE vs Polynomial Degree

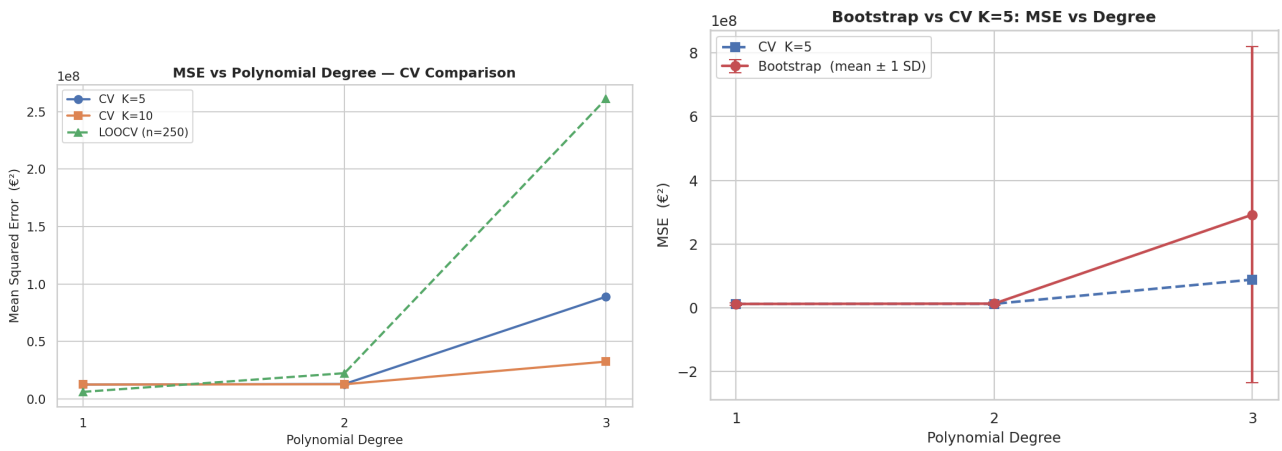

Figure 2: Left: CV MSE vs degree (K=5, K=10, LOOCV). Right: Bootstrap vs CV K=5. Both confirm degree 1 as optimal.

4. Task 2 — Model Selection

Task 2a: Subset Selection

Five criteria select the optimal number of predictors: **Cp = 6, AIC = 6, BIC = 3, Adj-R^2 = 7, CV-MSE = 4**. BIC is most parsimonious ($\ln(n) \approx 8.2$ penalty per parameter vs AIC's 2). Most consistently retained features: `duration`, `poutcome_success`, `contact_unknown`.

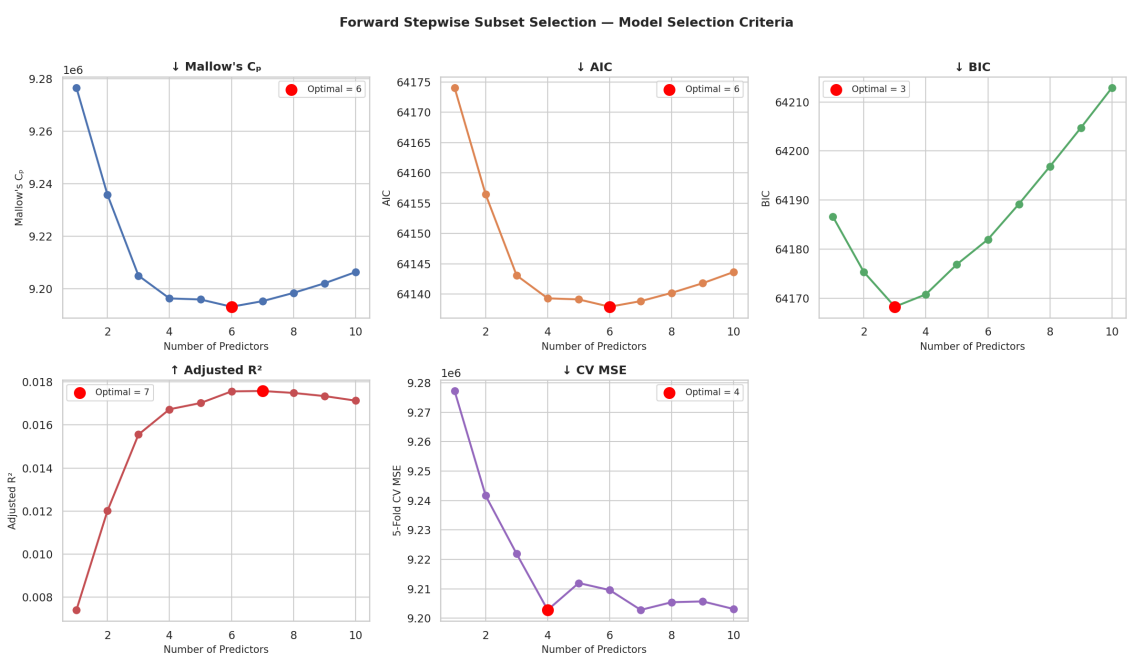

Figure 3: Forward stepwise selection: Cp, AIC, BIC, Adj-R^2, and CV-MSE vs number of predictors.

Task 2b: Shrinkage — Ridge and Lasso

CV-optimal: **Ridge λ = 322.46**, **Lasso λ = 7.61**. Ridge shrinks all coefficients smoothly; Lasso drives many to zero (sparsity). Bias-variance decomposition: bias^2 dominates total MSE; variance stays small. Ridge min-MSE at $\lambda = 0.0045$; Lasso at $\lambda = 1.3738$ (bias-variance optimal).

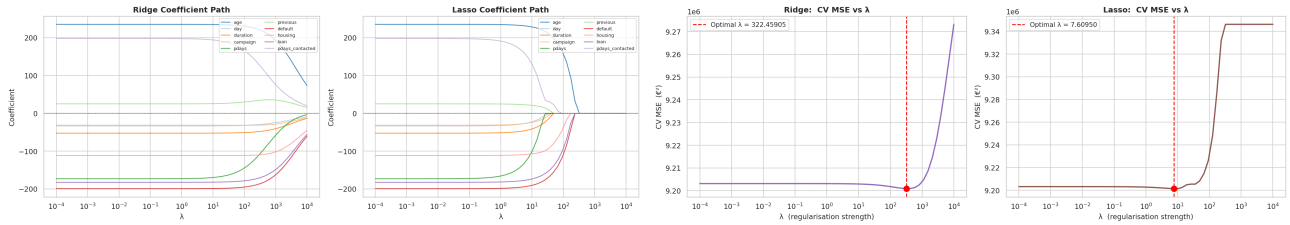

Figure 4: Left: Coefficient paths. Right: CV MSE vs λ (optimal marked red).

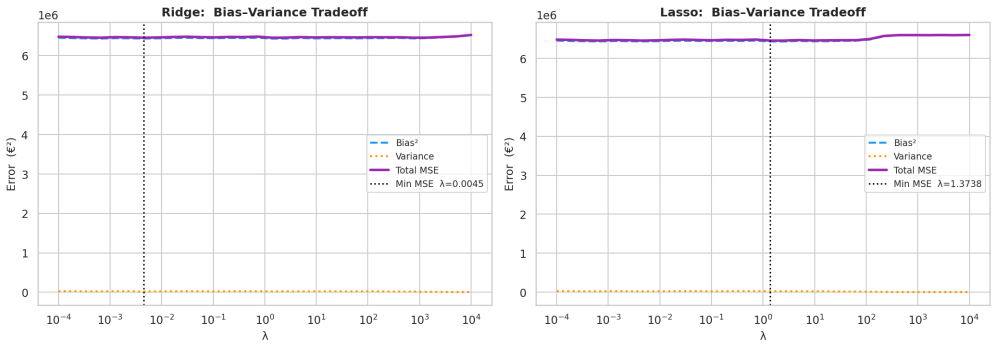

Figure 5: Bias-variance tradeoff for Ridge (left) and Lasso (right). Bias^2 dominates; variance is negligible.

Task 2c: PCR vs PLS

PC1 explains 24.3%, PC2 12.2% of variance. PCR minimum test MSE at **4 components**; PLS achieves comparable MSE with only **2 components** — supervised reduction is more efficient. Full OLS baseline MSE = 7,735,224.

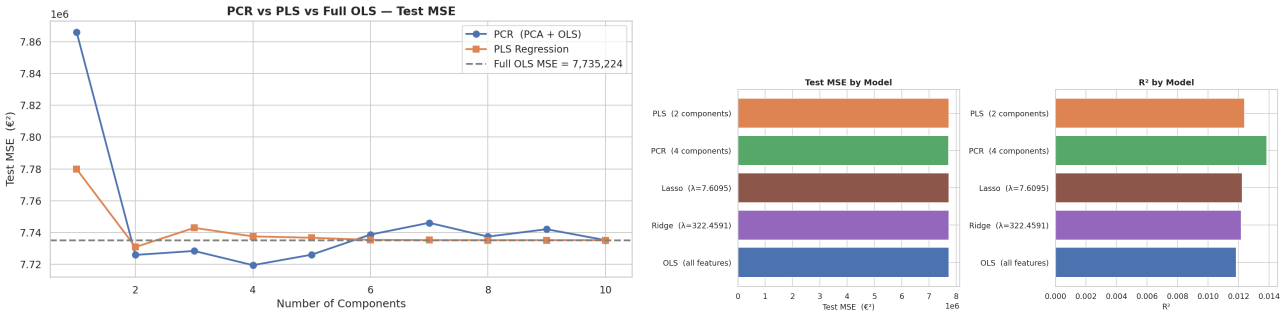

Figure 6: Left: PCR vs PLS test MSE. Right: Final comparison across all methods.

All methods yield similar test MSE ($~$7.5 \times 10^6$) and R^2 ($~$0.01–0.014). PLS (2 components) is most efficient; Lasso provides best interpretability; Ridge retains all features with reduced magnitude.

5. Conclusions

1. **LR is the best classifier** (AUC = 0.891); `poutcome_success` is the most actionable predictor.
2. **Youden's J threshold** = 0.027 raises recall to $~$90% for the minority class.
3. **`previous` is not a confounder** (0.07% coefficient change).
4. **Task 1:** Degree 1 is optimal; all CV and bootstrap methods agree.
5. **Task 2a:** BIC selects 3; Cp/AIC select 6; CV selects 4 features.

6. **Task 2b:** Ridge (λ = 322.46) and Lasso (λ = 7.61) both improve over OLS.
7. **Task 2c:** PLS (2 components) matches PCR (4 components), confirming supervised reduction is most efficient.
8. QDA underperforms LDA; campaign regression fails ($R^2 < 0$); no overfitting advantage from regularisation.

Team Contributions

Task	Tahar Masmaliyev	Abdulla Akhundzade
Data preprocessing, EDA, Classification	Lead	Support
Task 1: Resampling (CV & Bootstrap)	Lead	Support
Task 2: Model Selection (Subset, Shrinkage, PCR/PLS)	Support	Lead
Streamlit Dashboard and Report	Joint	Joint

References: Moro, S., Laureano, R., & Cortez, P. (2011). *Using Data Mining for Bank Direct Marketing.* ESM'2011, EUROSIS. James, G., Witten, D., Hastie, T., & Tibshirani, R. (2021). *An Introduction to Statistical Learning* (2nd ed.). Springer.